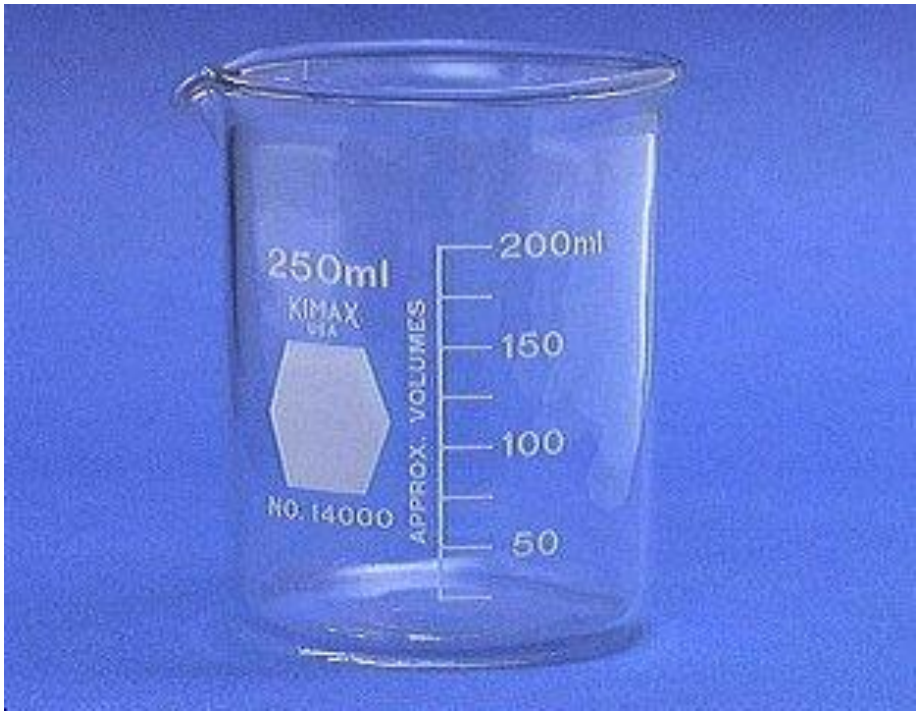
Lab Equipment

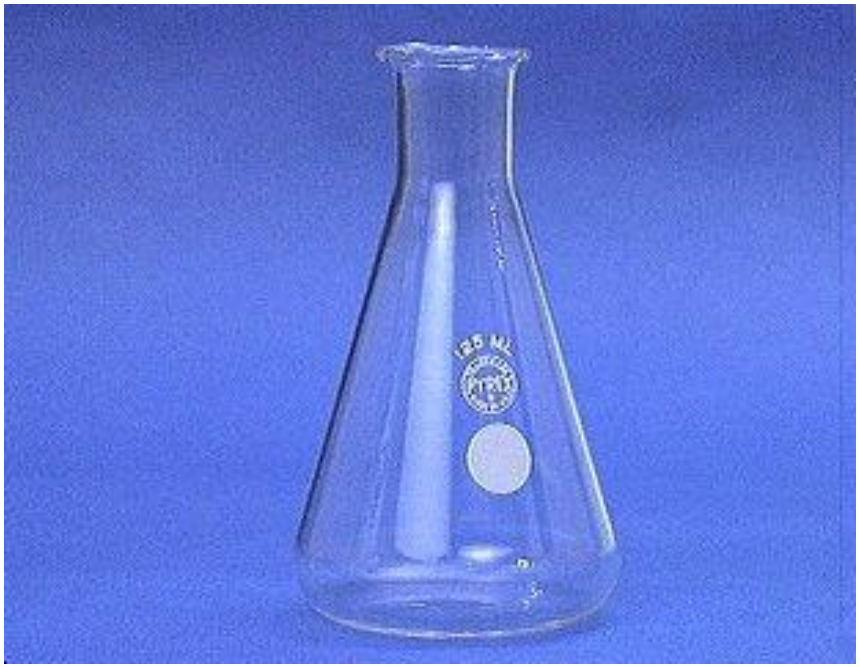


Beaker

Beakers hold solids or liquids that will not release gases when reacted or are unlikely to splatter if stirred or heated.



Erlenmeyer Flask



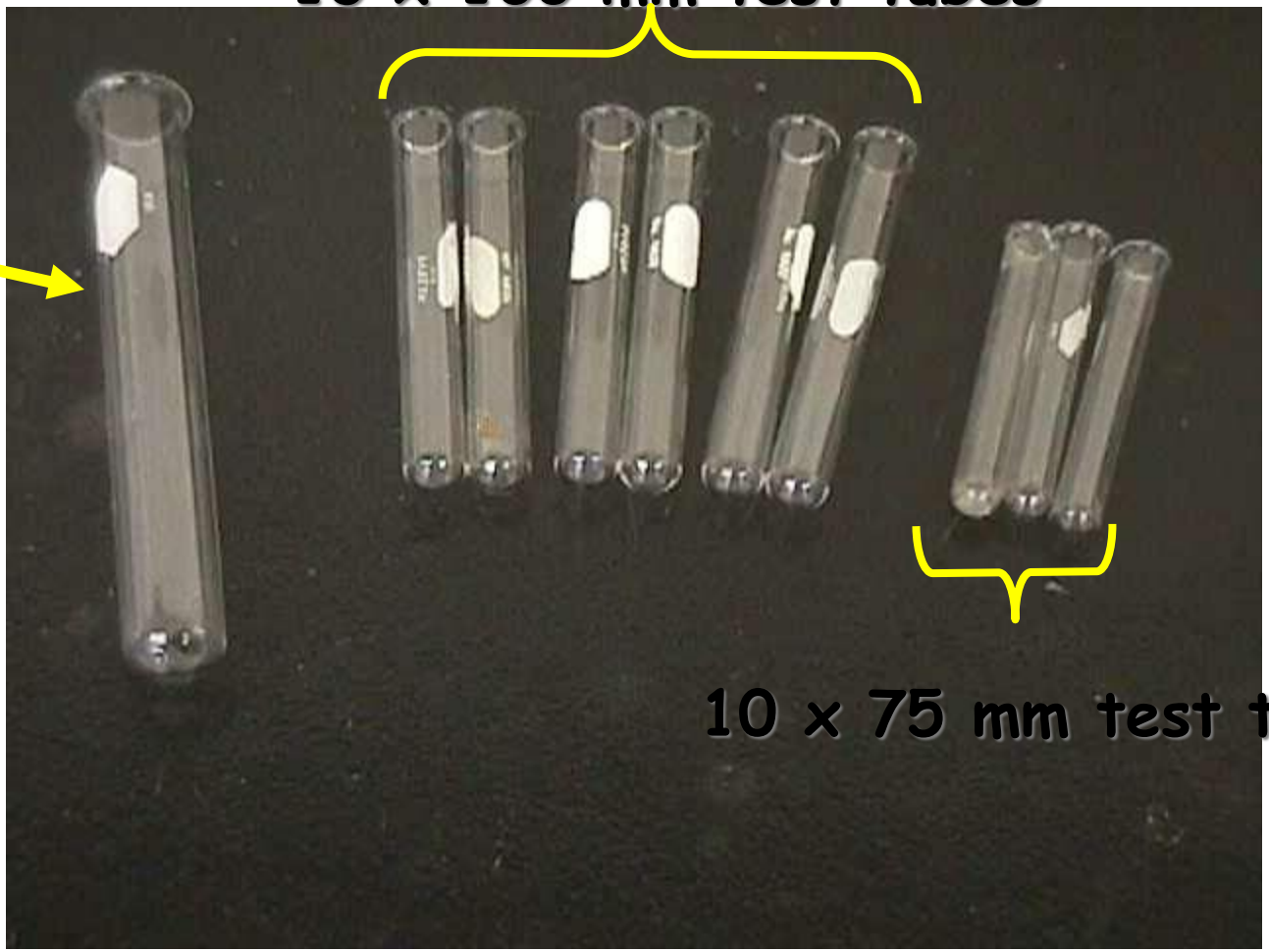
Erlenmeyer flasks hold solids or liquids that may release gases during a reaction or that are likely to splatter if stirred or heated.

Graduated Cylinder

A graduated cylinder is used to measure volumes of liquids.

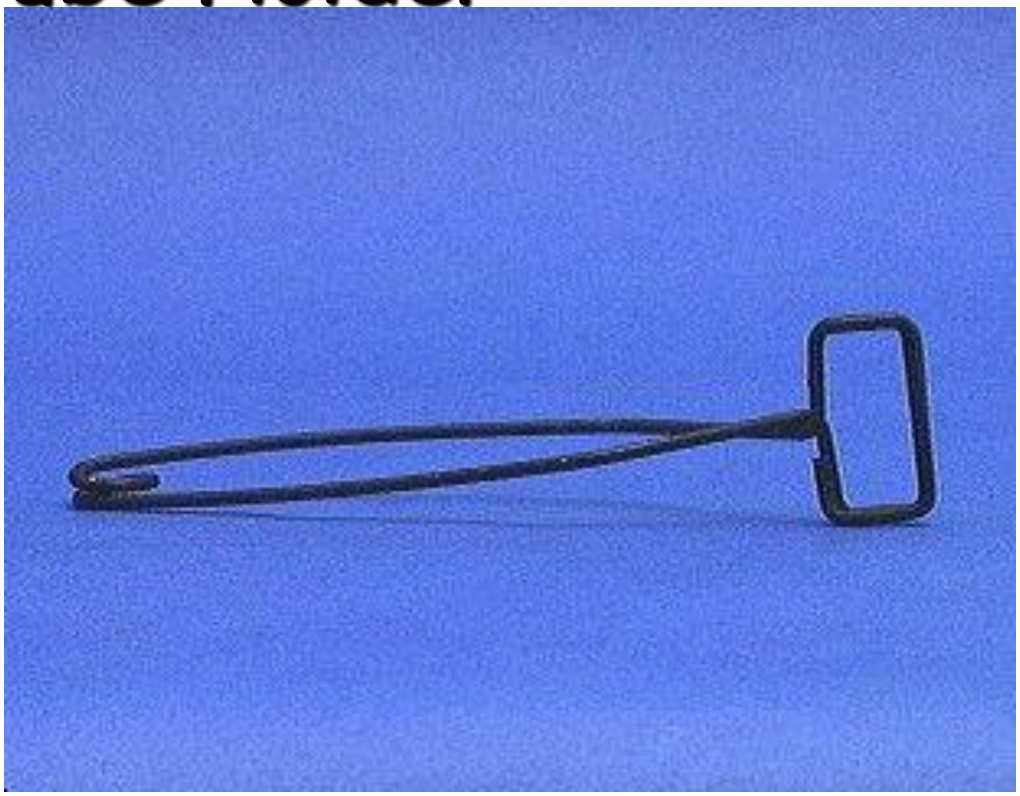


Test Tubes



Test Tube Holder

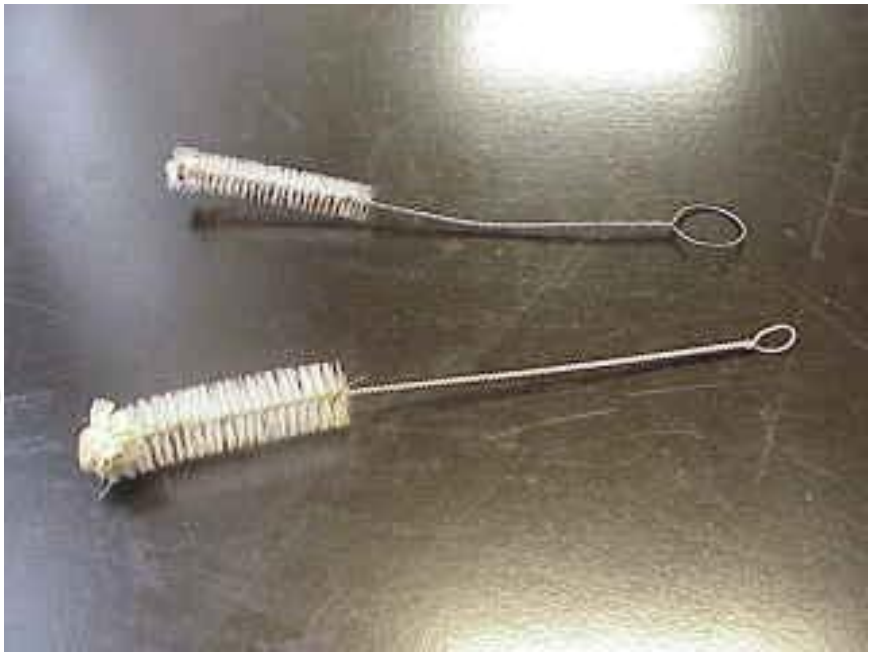
A test tube holder is useful for holding a test tube which is too hot to handle.



Test Tube Brushes

Test tube brushes are used to clean test tubes and graduated cylinders.

Forcing a large brush into a small test tube will often break the tube.



Test Tube Racks



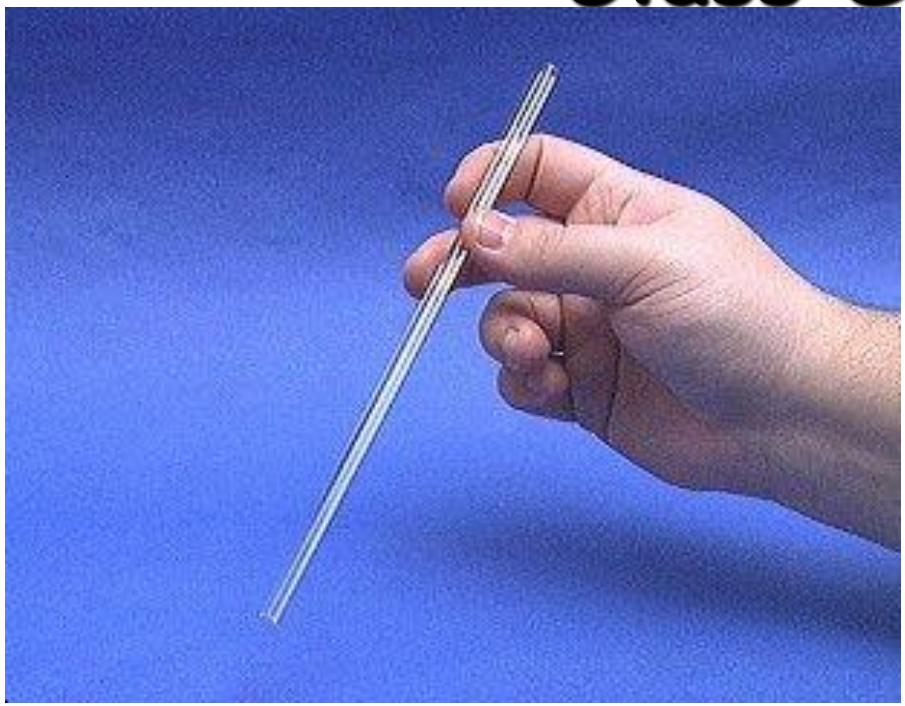
Test tube racks are for holding and organizing test tubes on the laboratory counter.
After washing flip test tube over on wooden peg to dry.

Spot Plates

Spot plates are used when we want to perform many small scale reactions at one time.



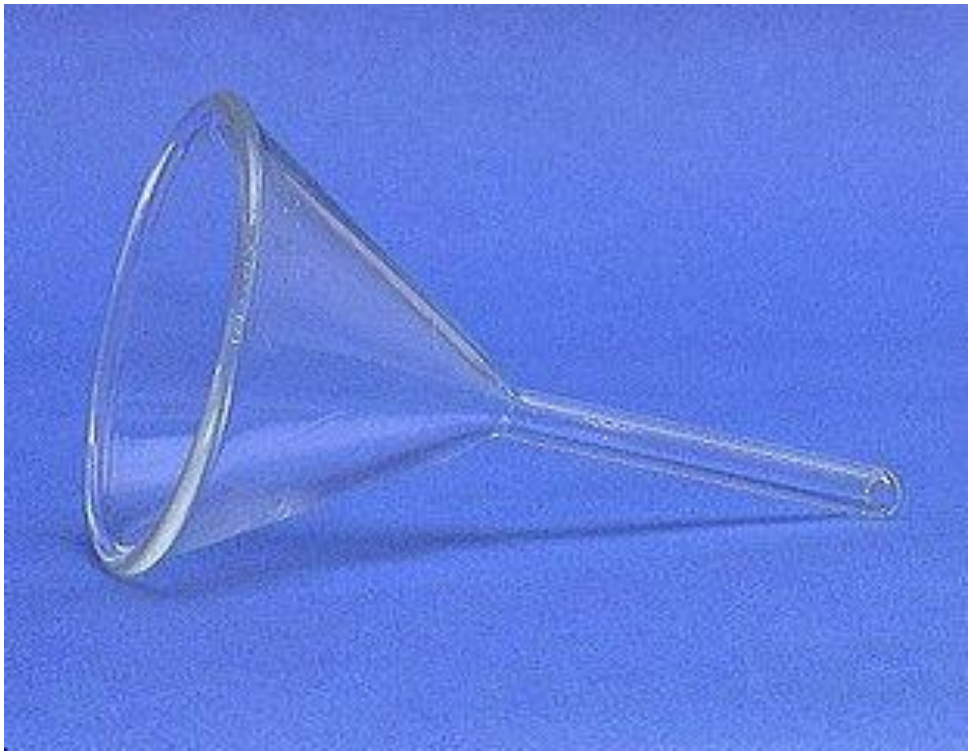
Glass Stir Rod



A glass rod is used to manually stir solutions. It can also be used to transfer a single drop of a solution.

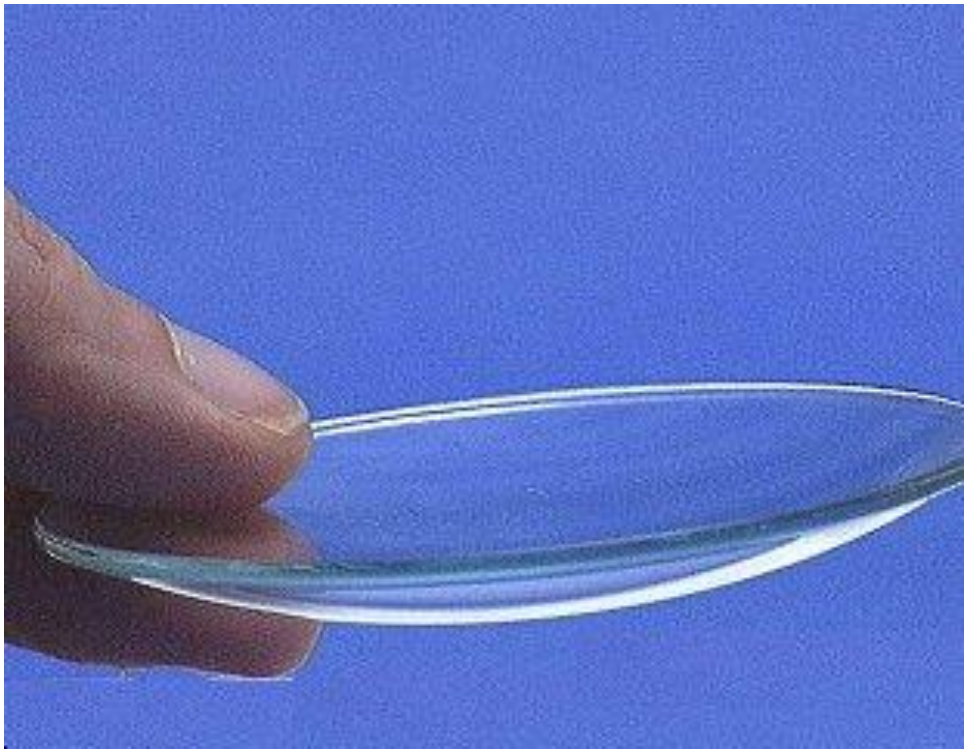
Funnel

A funnel is used to aid in the transfer of liquid from one vessel to another.



Watch Glass

A watch glass is used to hold a small amount of solid, such as the product of a reaction.



Wash Bottle



A wash bottle has a spout that delivers a wash solution to a specific area. Distilled water is the only liquid that should be used in a wash bottle.

Weighing Boat

Weighing boats are used to weigh solids that will be transferred to another vessel.



Bunsen Burner



Bunsen burners are used for the heating of nonvolatile liquids and solids.

Ringstands and their Components

Ringstand



Ringstands are a safe and convenient way to perform reactions that require heating using a Bunsen burner.

Ringstands and their Components

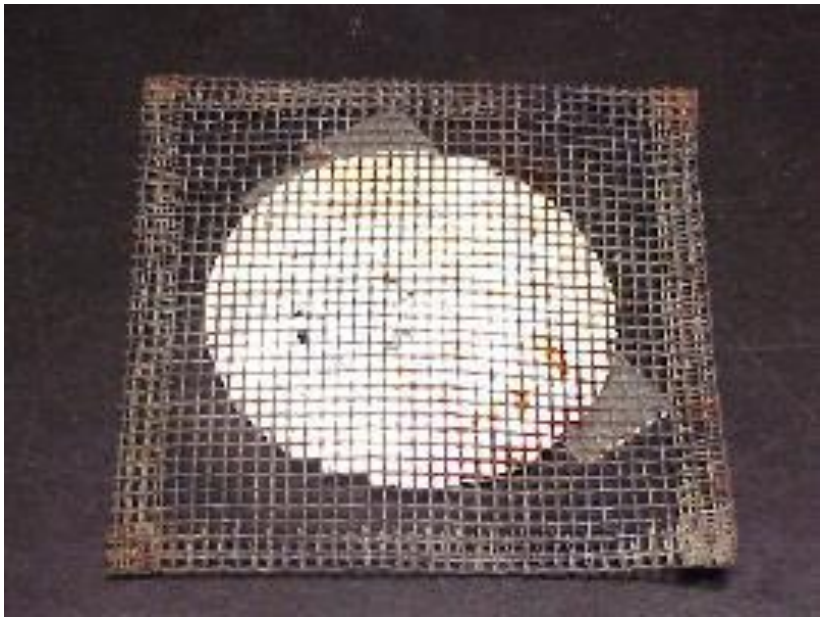
Utility Clamps

Utility clamps are used to secure test tubes, distillation columns, and burets to the ringstand.

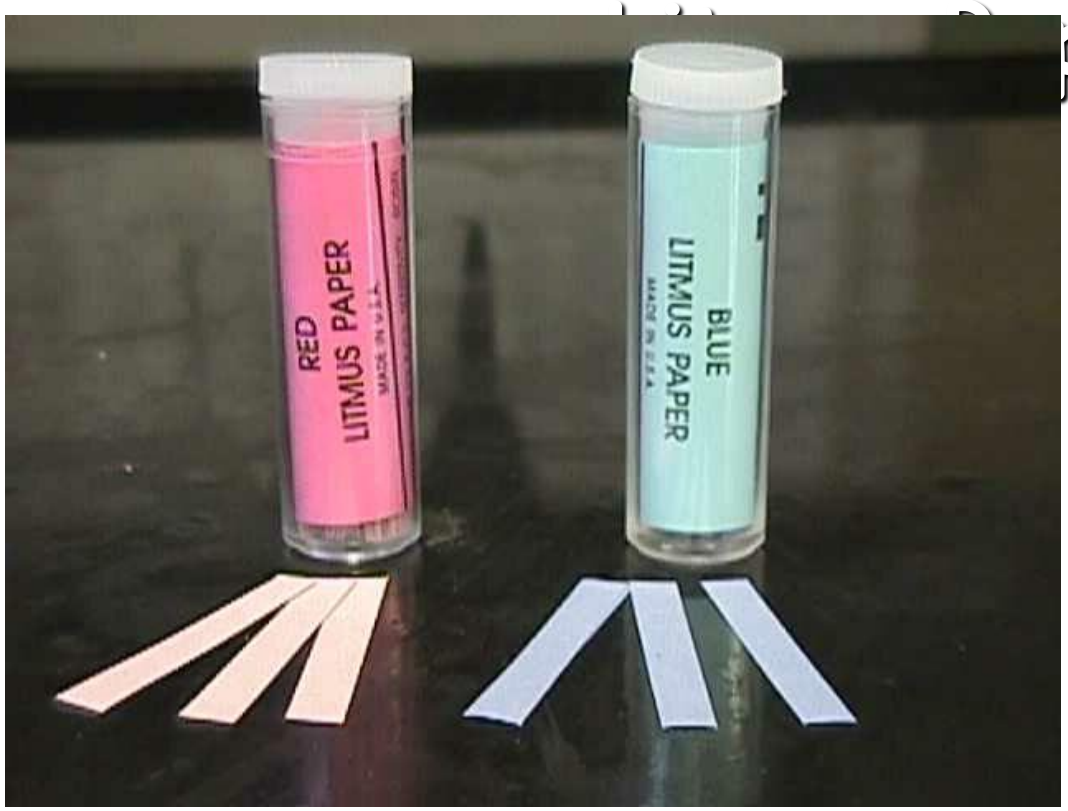


Ringstands and their Components
Wire Gauze

Wire gauze sits on the iron ring to provide a place to stand a beaker.



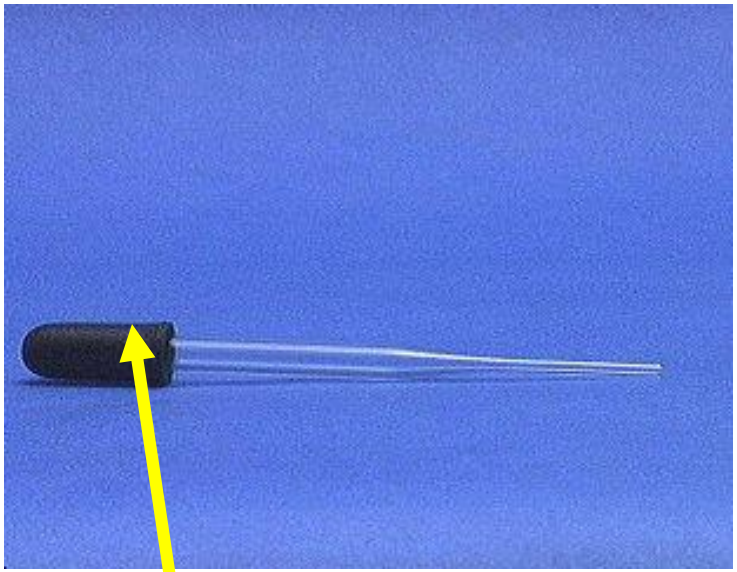
Litmus Paper



Red litmus paper is used to identify bases.

Blue litmus paper is used to identify acids.

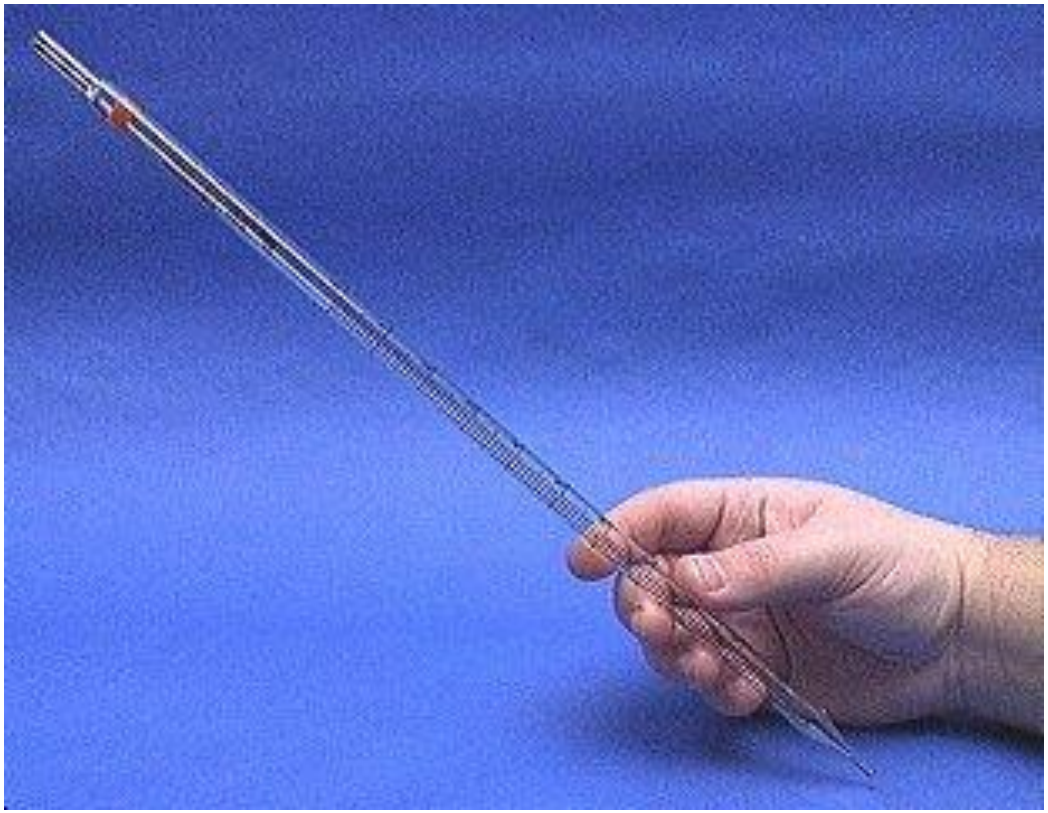
Medicine Dropper



A medicine dropper is used to transfer a small volume of liquid (less than one mL).

On top of each medicine dropper is a "rubber bulb"

Mohr Pipet

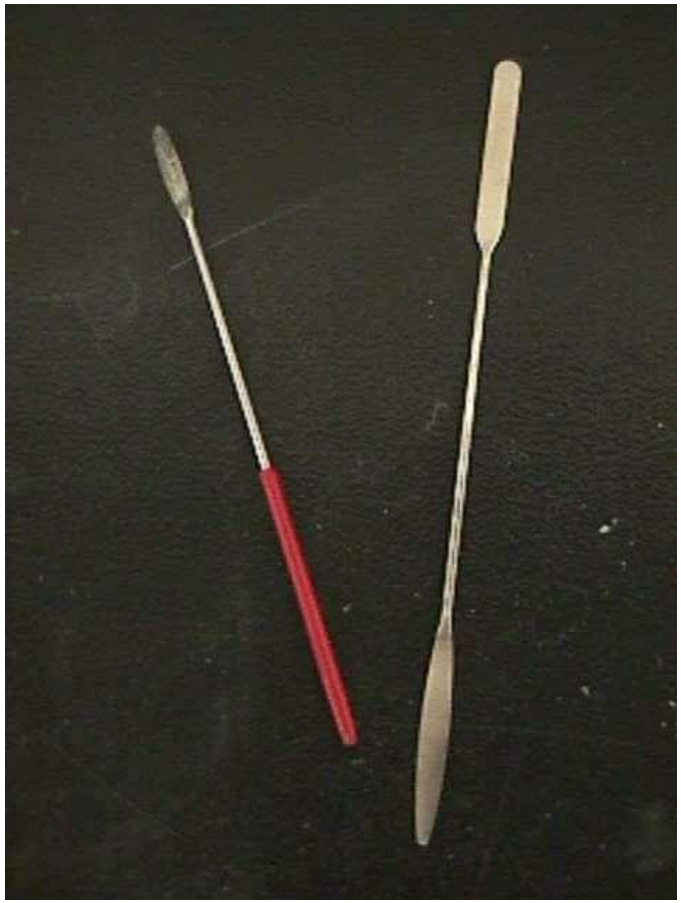


A Mohr pipet measures and delivers exact volumes of liquids.

Spatulas

Spatulas are used to dispense solid chemicals from their containers.

Chemicals should never be transferred with your bare hands.



Sensitive scale

The sensitive Scale provides a more accurate and precise reading and measurement of the weight of an object



Lab thermometer



Laboratory thermometers measure temperature with a high level of precision for applications such as experiment monitoring, instrument calibration, materials testing, and maintaining a sterile work environment.

A water bath

A water bath is a lab constant temperature equipment, providing heat source for varieties of devices that need heating. The circulating water bath is used to keep water at a constant temperature for incubating samples in a laboratory. Uses of Water bath: A laboratory water bath is used to heat samples.

A magnetic stirrer

A magnetic stirrer is a device widely used in laboratories and consists of a rotating magnet or a stationary electromagnet that creates a rotating magnetic field. This device is used to make a stir bar, immerse in a liquid, quickly spin, or stirring or mixing a solution

hot plates

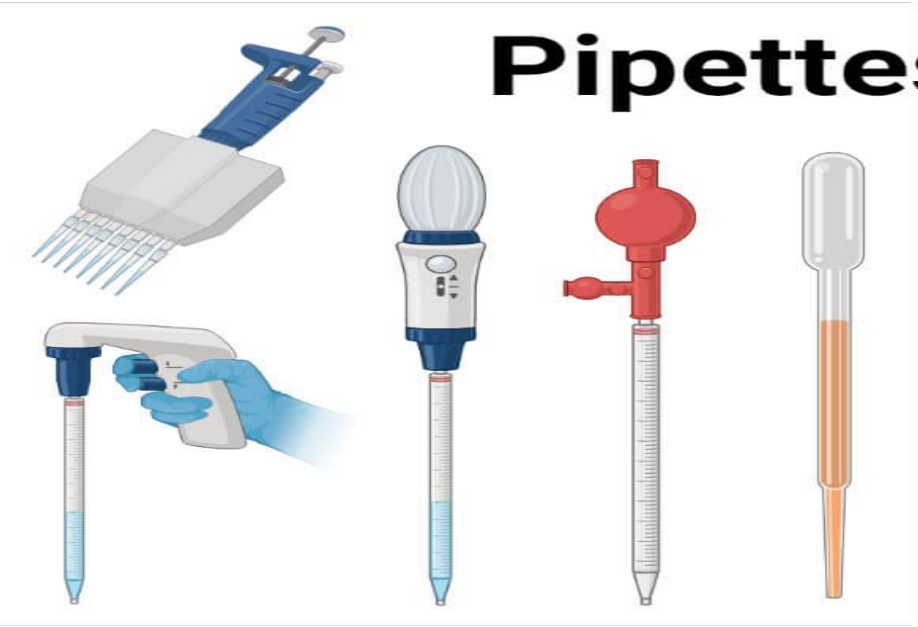


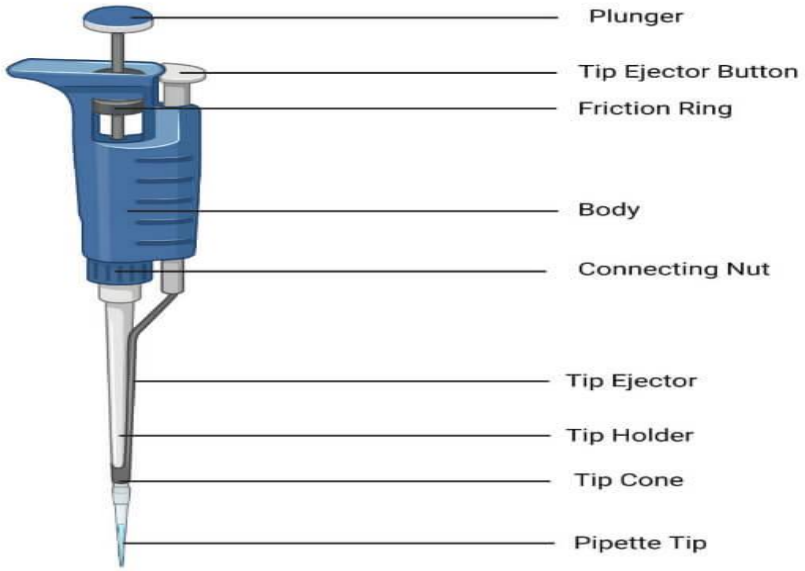
hot plates are frequently used in the laboratory to perform chemical reactions, to heat samples, and for numerous other activities. Hot plates are conceptually simple – a flat surface with heating elements. They do not produce open flames and are well suited for oil or sand bath use.

pH meter

A pH meter is an instrument used to measure hydrogen ion activity in solutions - in other words, this instrument measures acidity/alkalinity of a solution. The degree of hydrogen ion activity is ultimately expressed as pH level, which generally ranges from 1 to 14.







Examples of Pipette Usage

- Measuring small amounts of solvents and solutions, to collect and dispense liquids.
- In making serial dilutions, measuring / dispensing samples and diluted solutions.
- Measuring and dispensing media, reagents and cell cultures in biological experiments and laboratory tests.

Separating funnel

Separatory funnels are used in the lab for liquid-liquid extractions, separating a mixture's components into two solvent phases of different densities. The higher density liquid sinks to the bottom and can then be drained from a valve, leaving the less dense liquid in the funnel.

